Appendix for:

Learning Robot Visual Navigation in Crowds via Intention-Aware Scene Representations

A Robustness of Pose Detector under Various Perturbations

In this section, we present qualitative results to illustrate the performance of the adopted pose detector (Ultralytics YOLO-Pose[1]) across various scenarios in both simulated and real-world environments, as shown in Fig. 1. Specifically, our visualizations show that the detector can effectively estimate human poses under diverse pedestrian orientations relative to the robot, including frontal, lateral, and rear views when the pedestrian is fully visible. Furthermore, the model is able to yield reasonable estimations even in cases of mutual occlusions among pedestrians, or severe partial occlusions, such as when only the upper body or the head and shoulders are visible. These qualitative observations suggest that the pose detector can serve as a dependable perception module for our intention-aware navigation framework.

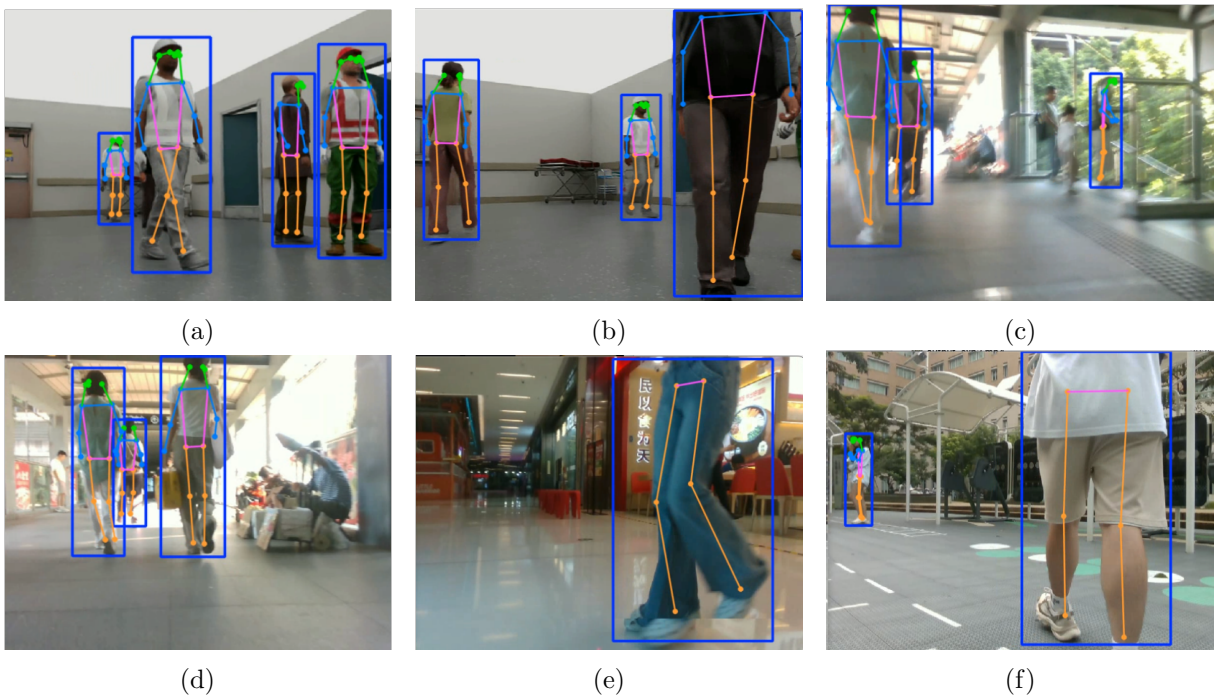

Figure 1: Qualitative evaluation of the pose detector across varied challenges. (a) Multiple pedestrian poses (frontal, lateral, rear). (b) Severely partial occlusion (above shoulders). (c-d) Mutual occlusions between people. (e-f) Significant occlusion (above waist). In all cases, the detector accurately captures the available keypoints.

In cases where specific keypoints are entirely undetected—such as under extreme truncation or heavy occlusion—we adopt a zero-padding strategy, following [2]. This approach leverages the inherent properties of the Transformer. Specifically, the self-attention mechanism naturally diminishes the influence of these zero-padded tokens. As a result, the model focuses on the available valid keypoints to infer pedestrian intentions.

B Detailed Quantitative Results with Standard Deviations

In this section, we provide the comprehensive quantitative results corresponding to the experiments presented in the main manuscript. Due to strict space constraints, the standard deviations

of the evaluation metrics across multiple evaluation episodes were omitted in the main text. Here, Table 1 and Table 2 report the detailed performance comparisons for the simulated crowd navigation and the long-horizon navigation tasks, respectively, including both the mean values and their corresponding standard deviations. These comprehensive metrics further demonstrate the stability and consistency of our proposed method across various environments and pedestrian densities.

Table 1: Comparison Results in Simulated Crowd Navigation

Density	**Method**	**Office Lobby**				**Hospital Corridor**				**Warehouse**			
		SR↑	NT↓	PL↓	TPZ↓	SR↑	NT↓	PL↓	TPZ↓	SR↑	NT↓	PL↓	TPZ↓
Low	**Ours**	**0.95**	9.03±0.42	**5.87±0.31**	**0.92±0.18**	**0.88**	9.82±0.58	6.19±0.29	**1.98±0.22**	**0.87**	9.77±0.61	**6.15±0.24**	**1.83±0.19**
	Ours (w/o I^2)	0.93	9.22±0.53	5.99±0.34	1.79±0.42	0.84	9.30±0.68	6.32±0.41	2.69±0.53	0.79	10.77±0.72	6.67±0.38	2.22±0.44
	Ours (w/o BEV)	0.92	9.15±0.48	5.95±0.32	1.57±0.34	0.81	9.97±0.74	6.97±0.43	2.54±0.47	0.74	10.85±0.69	6.71±0.42	2.12±0.38
	DRL-VO	0.93	**8.32±0.51**	5.91±0.38	1.29±0.43	0.80	**8.60±0.72**	**6.08±0.39**	3.08±0.62	0.80	**9.67±0.78**	6.38±0.44	2.33±0.52
	SARL*-OM	0.85	8.89±0.79	6.31±0.52	1.23±0.48	0.75	13.00±1.48	9.10±0.82	3.87±0.79	0.75	10.16±1.12	6.40±0.58	1.88±0.63
	ViNT	0.81	10.55±1.23	6.11±0.62	2.01±0.58	0.69	11.13±1.57	6.21±0.73	3.15±0.88	0.69	11.25±1.68	6.65±0.82	3.72±0.97
	DWA	0.92	12.48±1.27	6.24±0.58	1.66±0.62	0.87	11.34±1.42	6.01±0.63	2.57±0.78	0.77	13.16±1.58	6.71±0.74	3.42±0.89
High	**Ours**	**0.91**	9.20±0.58	6.18±0.32	**1.42±0.23**	**0.84**	9.83±0.72	6.31±0.38	**2.55±0.31**	**0.80**	11.15±0.82	7.02±0.43	**2.83±0.38**
	Ours (w/o I^2)	0.85	9.29±0.63	**5.89±0.28**	2.04±0.42	0.80	9.75±0.83	6.34±0.42	3.04±0.48	0.75	11.08±0.92	6.74±0.47	3.58±0.53
	Ours (w/o BEV)	0.83	9.31±0.57	6.14±0.33	1.65±0.38	0.77	9.91±0.78	6.67±0.44	2.89±0.52	0.73	11.45±0.87	6.97±0.48	3.39±0.47
	DRL-VO	0.83	**8.51±0.58**	**5.95±0.34**	2.54±0.48	0.72	**8.86±0.82**	**6.28±0.32**	3.01±0.57	0.71	**10.89±0.97**	**6.68±0.43**	3.71±0.64
	SARL*-OM	0.67	9.04±1.08	6.35±0.57	2.05±0.73	0.67	14.90±1.76	9.95±0.92	3.13±0.88	0.57	11.50±1.38	7.61±0.73	3.17±0.78
	ViNT	0.71	11.05±1.47	6.01±0.68	2.91±0.87	0.65	13.51±2.14	6.79±0.93	4.01±1.12	0.51	12.51±2.28	7.12±1.08	5.02±1.26
	DWA	0.75	12.94±1.56	6.86±0.73	2.88±0.78	0.75	12.46±1.82	6.35±0.68	3.50±0.92	0.72	15.18±2.17	7.60±0.88	4.79±1.15

Table 2: Long-Horizon Comparison Results

Env	Method	SR↑	NT↓	PL↓	TPZ↓
Office	**Ours**	**0.95**	**40.82±1.52**	26.53±0.85	**1.70±0.25**
	Ours (w/o I^2)	0.80	49.39±2.10	**24.21±0.78**	2.10±0.30
	Ours (w/o BEV)	0.83	47.01±1.95	28.21±0.90	2.20±0.35
	DRL-VO	0.75	41.02±1.80	25.02±0.82	1.80±0.28
	SARL*-OM	0.33	68.64±4.50	39.81±2.10	7.96±1.05
	ViNT	0.41	55.87±3.80	32.49±1.60	5.61±0.85
	DWA	0.37	61.21±4.10	27.32±1.10	3.80±0.60
Hospital	**Ours**	**0.79**	40.89±1.60	22.49±0.75	**0.42±0.15**
	Ours (w/o I^2)	0.72	37.15±1.45	**20.06±0.68**	1.37±0.25
	Ours (w/o BEV)	0.73	42.34±1.85	23.29±0.80	1.21±0.22
	DRL-VO	0.67	**35.31±1.40**	21.90±0.70	1.48±0.28
	SARL*-OM	0.42	46.90±2.20	30.02±1.15	1.65±0.35
	ViNT	0.56	42.51±1.90	24.38±0.95	1.89±0.40
	DWA	0.50	45.19±2.15	25.25±1.05	2.15±0.45

References

[1] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics yolov8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[2] T. Salzmann *et al.*, "Robots that can see: Leveraging human pose for trajectory prediction," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7090–7097, 2023.